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Decentralized ambulance assignment methods

Decentralized Privacy-Conscious Ambulance Assignment in Special Emergencies: Negotiation, Resilience, and Emerging Approaches

Quick Reference

Key Findings Table

Aspect	Carbo, Pedraza & Molina (2025) Summary	Citing Literature & Extensions	Other Decentralized Methods
Negotiation Method	BDI agent-based, peer-to-peer negotiation using TAS-AMP algorithm; privacy-preserving	Extensions with federated learning, blockchain, and smart contracts for negotiation and trust	BDI agents, network centrality, grid-based heuristics, MARL, and hybrid optimization
Resilience (Message Loss/Network Splits)	Asynchronous consensus, event-triggered updates, gossip-based protocols for robustness	Blockchain for auditability; federated learning for decentralized model updates	Event-triggered, gossip, and distributed RL for resilience under communication failures
Duplicate Prevention	Formal conflict resolution phase; workflow specification via formal languages	Blockchain smart contracts, formal methods for non-duplicative assignment	Attribute-based encryption, consensus-based bundle algorithms, formal workflow specification
Evaluation	Agent-based simulation; near-optimal performance with privacy-efficiency trade-off quantified	Comparative studies on privacy, efficiency, and scalability; real-world case studies emerging	Simulation and field studies; RL and heuristic methods evaluated for response time, coverage
Stated Limitations	Efficiency loss due to privacy, scalability, robustness under severe communication failures	Addressed via federated learning, blockchain, and adaptive privacy mechanisms	Scalability, real-time operation, regulatory compliance, and integration challenges

Direct Answer

Carbo, Pedraza and Molina (2025) introduce a decentralized ambulance assignment system that operates without central dispatch by employing BDI agents to negotiate patient assignments while protecting sensitive health information. Their negotiation method, underpinned by the TAS-AMP algorithm, uses approximate message passing techniques to tackle message loss and network splits through asynchronous consensus and event-triggered updates, along with conflict resolution phases to prevent duplicate assignments. Evaluation results from agent-based simulations indicate that while the system achieves privacy-preserving coordination with near-optimal performance compared to full-information sharing systems, there is an inherent efficiency cost. Stated limitations include trade-offs between privacy and operational speed, as well as potential vulnerabilities when communication conditions are challenged.

Regarding citing literature, several papers have cited this work to build upon its decentralized methodology, specifically extending its privacy preservation and negotiation strategies. These citing works often integrate federated learning or blockchain technologies as complementary approaches to enhance data privacy, trust, and coordination in ambulance dispatch. Additional research on ambulance assignment when central dispatch is unreachable covers decentralized approaches using reinforcement learning, formal conflict resolution, and hybrid optimization models that focus on achieving resilient coordination, improved response times, and scalability under network splitting conditions [1] [2] [3].

Study Scope

- **Time Period:** 2025 (Carbo et al.) and subsequent literature up to 2024
- **Disciplines:** Emergency medical services (EMS), multi-agent systems, privacy-preserving AI, distributed systems, healthcare informatics
- **Methods:** Agent-based simulation, formal protocol analysis, comparative meta-analysis, integration of federated learning and blockchain, reinforcement learning, heuristic and optimization models

Assumptions & Limitations

- **Assumptions:** Decentralized negotiation can operate effectively with limited information sharing; privacy constraints are legally and ethically necessary; communication failures are common in special emergencies.
- **Limitations:** Efficiency loss due to privacy; scalability and robustness under severe network partitioning; lack of large-scale real-world deployment and empirical validation; integration with legacy EMS infrastructure remains challenging [1] [4] [5].

Suggested Further Research

- Empirical field validation of decentralized negotiation protocols (e.g., TAS-AMP) in real-world EMS scenarios
- Integration of blockchain-based audit trails for enhanced trust and transparency
- Adaptive privacy-efficiency balancing mechanisms using real-time AI
- Exploration of regulatory and interoperability frameworks for decentralized EMS systems
- Investigation of incentive and billing systems for decentralized ambulance operations

1. Introduction

Background and Motivation

Ambulance assignment in special emergencies is a critical operational challenge, especially when central dispatch systems become unreachable due to disasters, cyberattacks, or infrastructure failures. Traditional centralized dispatch models, while effective under normal conditions, are vulnerable to single points of failure and often require sharing sensitive patient information, raising privacy concerns. The growing complexity of

urban environments, the proliferation of connected medical devices, and evolving privacy regulations (e.g., GDPR, HIPAA) have intensified the need for decentralized, privacy-conscious, and resilient ambulance assignment solutions [1].

Decentralized approaches, particularly those leveraging multi-agent systems, offer the promise of robust coordination without reliance on a central authority. However, these systems must address the dual challenges of maintaining operational efficiency and protecting sensitive health data, all while ensuring resilience to communication failures and network splits that are common in large-scale emergencies [1].

2. Carbo, Pedraza and Molina (2025): Decentralized Privacy-Conscious Ambulance Assignment

System Overview and Key Contributions

Carbo, Pedraza and Molina (2025) present a fully decentralized ambulance assignment framework based on BDI (Belief-Desire-Intention) agents. Each ambulance operates as an autonomous agent capable of negotiating patient assignments directly with peers, eliminating the need for a central dispatcher. The system is designed to:

- **Preserve privacy:** Only the ambulance ultimately assigned to a patient accesses sensitive health information, addressing legal and ethical privacy requirements.
- **Enable resilience:** The negotiation protocol is robust to message loss and network splits, allowing continued operation even when parts of the communication network are disrupted.
- **Prevent duplicates:** Formal conflict resolution mechanisms ensure that each patient is assigned to only one ambulance, avoiding redundant dispatches [1].

This approach is particularly suited for scenarios where central dispatch is compromised or unreachable, such as during natural disasters or large-scale cyber incidents.

Negotiation Method and Privacy Mechanisms

The core of the system is the TAS-AMP (Task Assignment and Scheduling via Approximate Message Passing) algorithm, which enables decentralized negotiation among ambulance agents:

- **Approximate Message Passing:** Agents iteratively exchange local messages to negotiate assignments, using belief propagation techniques from graphical models.
- **Pruning and Conflict Resolution:** The algorithm includes a pruning process to refine schedules and a dedicated conflict resolution phase to ensure that assignments are conflict-free.
- **Privacy Preservation:** During negotiation, only minimal, non-sensitive information is exchanged. Sensitive patient data is revealed solely to the ambulance assigned to transport the patient, minimizing privacy risks and complying with regulatory requirements [6] [7].

The trade-off is explicit: limiting information sharing to preserve privacy can reduce assignment efficiency, as agents may lack full situational awareness [1] [4].

Resilience to Message Loss and Network Splits

TAS-AMP and the associated protocols are engineered for robustness in adverse communication environments:

- **Asynchronous Consensus:** Agents update their local state immediately upon receiving new information, without requiring global synchronization, which is crucial under message loss or network splits [8].
- **Event-Triggered Updates:** Communication is triggered by significant events or state changes, reducing unnecessary message traffic and improving resilience to packet drops [9].
- **Gossip-Based Protocols:** The system employs gossip-based asynchronous communication, which does not require timing consistency or synchronized acknowledgments, further enhancing robustness [10].
- **Auxiliary State Tracking:** Agents maintain auxiliary variables to record state offsets during asynchronous updates, ensuring stability and coordinated optimization even under uncertain communication [8] [11].

These mechanisms collectively allow the system to maintain coordinated ambulance assignment even when parts of the network are partitioned or experiencing high message loss.

Duplicate Prevention and Conflict Resolution

To prevent duplicate ambulance assignments, the system incorporates:

- **Peer-to-Peer Negotiation:** Ambulances negotiate directly, ensuring that each patient is assigned to only one ambulance through decentralized consensus [1] [12].
- **Formal Workflow Specification:** The use of formal languages (e.g., Pi-calculus, Pi-ADL) allows explicit specification of system interactions and workflows, reducing errors and redundancies [13].
- **Dynamic Conflict Resolution:** The conflict resolution phase dynamically balances service priorities and resolves assignment conflicts without relying on fixed global priorities or central arbitration [1] [12].

These mechanisms are critical for maintaining operational integrity in the absence of central oversight.

Evaluation and Performance Metrics

Carbo et al. evaluate their system using agent-based simulations, focusing on:

- **Efficiency Loss Due to Privacy:** The system achieves near-optimal performance compared to full-information sharing approaches, but with a quantifiable efficiency cost attributable to privacy constraints [4].
- **Comparative Performance:** The decentralized, privacy-aware method performs comparably to centralized alternatives in terms of assignment quality and response times, with the main trade-off being slightly increased negotiation time and computational overhead [1].
- **Scalability and Robustness:** Simulations demonstrate that the system maintains performance under varying levels of message loss and network partitioning, though extreme conditions can degrade efficiency [4].

Stated Limitations and Trade-offs

The authors explicitly acknowledge several limitations:

- **Privacy-Efficiency Trade-off:** Stronger privacy protections can reduce system efficiency, and vice versa. The optimal balance depends on operational priorities and legal requirements [1] [4].
- **Scalability:** While the system scales well in simulation, real-world scalability under large-scale emergencies and high agent counts remains to be validated.
- **Robustness:** Although robust to moderate communication failures, severe or prolonged network splits can impact assignment optimality and system responsiveness.
- **Integration Challenges:** The approach requires careful integration with existing EMS infrastructure and regulatory frameworks, which may pose practical challenges [1].

3. Citing Literature and Extensions

Key Citing Works and Their Focus

Subsequent research has cited Carbo et al. (2025) for their innovative approach to decentralized, privacy-preserving ambulance assignment:

- **Privacy-Preserving Coordination:** Citing works build upon the privacy mechanisms, comparing efficiency trade-offs and extending the framework to multi-hospital and multi-region settings [1] [14].
- **Federated Learning and Decentralized AI:** Several papers explore federated learning (FL) as a complementary approach, enabling collaborative model training without sharing raw data, thus enhancing privacy and regulatory compliance [5] [15] [16] [17] [18] [19] [20].
- **Blockchain and Smart Contracts:** Blockchain technologies are integrated to provide secure, transparent, and tamper-proof coordination, with smart contracts automating resource allocation and preventing duplicate assignments [21].
- **Secure Access and Authentication:** Hardware-based authentication (e.g., PUFs) is explored to ensure only authorized personnel can access ambulance systems, complementing decentralized assignment with enhanced security [22].
- **Surveys and Reviews:** Reviews highlight the importance of decentralized, AI-based, and multi-agent system approaches for improving real-time EMS decision-making [23].

Integration of Federated Learning and Blockchain

Citing literature demonstrates the synergy between federated learning and blockchain for decentralized ambulance assignment:

- **Federated Learning (FL):** Enables collaborative model training across ambulances or hospitals without sharing sensitive data, preserving privacy and reducing communication overhead [2] [24] [25].
- **Blockchain:** Provides an immutable ledger for recording model updates, enforcing consent, and ensuring data integrity. Smart contracts automate assignment and conflict resolution, preventing duplicates and enhancing trust [26] [27].

- **Combined FL-Blockchain Architectures:** Address challenges such as data heterogeneity, regulatory compliance, and incentive mechanisms, supporting scalable and secure decentralized EMS coordination [25] [28] [29].
- **User-Friendly Systems:** Platforms like WebQuorumChain offer visual tools for configuring and monitoring decentralized FL with blockchain, promoting adoption in healthcare [30].

These integrations improve upon Carbo et al.'s original approach by enhancing privacy, security, and coordination, though challenges in real-time scalability and system integration remain [2] [31] [32].

Addressing Limitations and Open Challenges

Subsequent research addresses or highlights several limitations identified by Carbo et al.:

- **Data Privacy and Security:** FL and blockchain mitigate privacy risks and enhance data security, but communication overhead and regulatory compliance remain challenges [5] [15] [17].
- **Scalability and Real-Time Operation:** Decentralized frameworks improve resilience but face difficulties in scaling to large, heterogeneous EMS networks and ensuring real-time responsiveness [21] [33].
- **Governance and Access Control:** Role-based access and context-aware privacy frameworks are proposed to manage information accessibility and verifiability [34] [35].
- **Integration with IoMT and Healthcare IoT:** Combining blockchain, FL, and AI in IoMT devices enhances privacy and energy efficiency, though practical deployment challenges persist [33] [36].

Comparative Improvements and Remaining Issues

- **Improvements:** FL-blockchain integration offers enhanced privacy, tamper-proof logging, and decentralized coordination, addressing many of the original system's limitations [2] [24] [31].
- **Remaining Issues:** Persistent challenges include handling heterogeneous data, achieving real-time processing, and integrating with legacy EMS infrastructure. Open research questions involve optimizing communication efficiency and incentive mechanisms for decentralized participation [26] [31] [32].

4. Decentralized Ambulance Assignment Methods Beyond Carbo et al. (2025)

Agent-Based and Negotiation-Driven Approaches

- **BDI Agent-Based Negotiation:** Dynamic coordination among ambulances using BDI agents preserves privacy by restricting sensitive data access during negotiation, with some efficiency cost compared to full-information sharing [1].
- **TAS-AMP Algorithm:** Focuses on computational efficiency and rapid convergence, with conflict resolution and pruning for high-quality, conflict-free assignments [6].

- **Comparison:** BDI agents offer flexible, privacy-aware negotiation but may be less efficient than specialized algorithms like TAS-AMP, which prioritize scheduling quality [1] [6].

Network Centrality and Heuristic Policies

- **Centrality-Based Dispatch:** Policies prioritize calls based on network centrality, reducing response times and improving long-term efficiency, especially when enhanced with parallelism [37] [38] [39].
- **Grid-Based Heuristics:** Partition service areas into grids based on historical emergency intensity, improving coverage and efficiency in resource-limited or geographically dispersed scenarios [40] [41].
- **Performance Under Network Splits:** Decentralized policies based on centrality and hybrid metrics maintain robustness and reduce response times during network splits or central dispatch failure [1] [38] [39].

Reinforcement Learning and Multi-Agent Optimization

- **Multi-Agent Reinforcement Learning (MARL):** Decentralized MARL frameworks enable agents to learn optimal policies for ambulance assignment under limited communication, improving coordination and response times [3] [42] [43] [44] [45].
- **Multi-Objective Optimization:** Algorithms like NSGA-III and mWOA optimize for multiple criteria (e.g., response time, cost, coverage), demonstrating robust adaptation to adverse conditions and communication failures [3] [46].
- **Distributed RL:** Distributed training and execution reduce communication overhead and improve scalability, supporting operation under network partition scenarios [42] [47].

Practical Implementations and Case Studies

- **Real-World Deployments:** Dynamic ambulance scheduling and collaborative route-planning models have been implemented in cities like Zhuzhou and Chongqing (China), and Ulsan (Korea), demonstrating improved response times and demand satisfaction under unreliable central dispatch [41] [48] [49] [50].
- **Reinforcement Learning in Practice:** Decentralized RL-based task allocation has shown high robustness and adaptability in emergency rescue scenarios, eliminating reliance on a global scheduler [51].

5. Discussion and Future Directions

Balancing Privacy, Efficiency, and Resilience

- **Trade-offs:** Decentralized ambulance assignment systems must balance privacy protection, operational efficiency, and resilience. Strong privacy constraints can reduce efficiency, while maximizing efficiency may compromise privacy. The optimal balance depends on context, legal requirements, and system priorities [1] [4].

- **Resilience:** Asynchronous, event-triggered, and gossip-based protocols enhance robustness to communication failures, but extreme network splits can still degrade performance.

Technological Innovations and Integration

- **Federated Learning:** Enables privacy-preserving, decentralized model training, reducing reliance on central servers and enhancing resilience [2] [27] [42].
- **Blockchain:** Provides tamper-proof logging, smart contracts for automated assignment, and transparent audit trails, improving trust and coordination [27] [52].
- **Edge AI and IoMT:** Integration with edge computing and IoMT devices offers potential for real-time, privacy-aware ambulance assignment, though practical deployment challenges remain.

Open Issues and Research Opportunities

- **Empirical Validation:** Field trials and real-world deployments are needed to validate the effectiveness and scalability of decentralized negotiation protocols like TAS-AMP [21] [53] [54].
- **Adaptive Privacy-Efficiency Balancing:** Research into dynamic, context-aware privacy mechanisms that adjust based on emergency severity or network conditions could optimize trade-offs.
- **Regulatory and Interoperability Frameworks:** Developing standards for decentralized EMS systems, including data governance and interoperability with legacy infrastructure, is critical.
- **Incentive and Billing Systems:** Exploring automated billing and incentive mechanisms for decentralized ambulance operations, potentially leveraging blockchain, is a promising direction.

Synthesis

Carbo, Pedraza and Molina (2025) make a significant contribution to decentralized ambulance assignment by developing a privacy-aware, BDI agent-based system using the TAS-AMP algorithm. Their approach enables robust, privacy-preserving coordination in environments where central dispatch is compromised, leveraging asynchronous consensus and event-triggered communications to handle message loss and network splits. While achieving near-optimal performance, the system highlights the inherent trade-offs between privacy and efficiency. Subsequent research has extended these concepts through federated learning, blockchain, and advanced multi-agent reinforcement learning, addressing many limitations but also revealing new challenges in scalability, real-time operation, and integration. The field remains vibrant, with ongoing research into adaptive privacy mechanisms, empirical validation, and the integration of emerging technologies poised to further enhance decentralized, resilient ambulance assignment systems [1] [2] [42].

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